

Total number of printed pages-3

02-23R/2024 (MAT 2103 SE)

2024

MATHEMATICS

(Skill Enhancement)

Paper : MAT 2103 SE

(Latex and HTML)

Full Marks : 35

Time : 1½ hours

The figures in the margin indicate full marks for the questions.

1. Answer the following questions : $1 \times 4 = 4$

(a) What does HTML stand for ?

(b) What is the purpose of the **
** tag in HTML ?

(c) Name the LaTeX package used for handling images.

(d) What is the command in LaTeX to start the document environment ?

2. Answer to the following : $2 \times 3 = 6$

(a) Explain the function of the **<head>** and **<body>** tags in an HTML document.

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(b) What is the use of the **href** attribute in HTML? Provide an example.

(c) Explain the use of the **\section** command in a LaTeX document.

3. Answer **any three** of the following:

5×3=15

(a) Describe how to include and style an image in an HTML document using the **src**, **alt**, and **style** attributes.

(b) List and describe various HTML formatting elements like ****, ****, **<i>** and **** and their importance in web design.

(c) Describe how to use the **style** attribute in HTML to change text color and font size. Provide examples.

(d) Discuss the LaTeX **table** environment and demonstrate how to create a simple three-column table.

(e) Write the LaTeX syntax to display the

expression
$$\frac{\sqrt{x^2 + y^2}}{x + y}$$

Include instructions on how to manage the square root and the fraction to ensure clarity in the representation of the numerator and denominator.

4. Answer **any one** of the following: 10

- (a) Create a LaTeX table similar to the one in the image, but extend it to have three columns and three rows in the body of the table. Label the additional row as 'r3' and the additional column as 'c3'. Use 'e', 'f', 'g', 'h', and 'i' for the new cells' entries. Ensure that the headers 'R' and 'C' still span the correct number of rows and columns respectively. Provide the complete LaTeX source code for your table.

R	C		
	c1	c2	c3
r1	a	b	h
r2	c	d	h

- (b) Write a detailed explanation of how HTML elements are structured, including examples of nested elements and their impact on webpage layout.

e	c ₂	c ₃
M ₁		
M ₂		

	c ₁	c ₂	c ₃
M ₁			
M ₂			
M ₃			